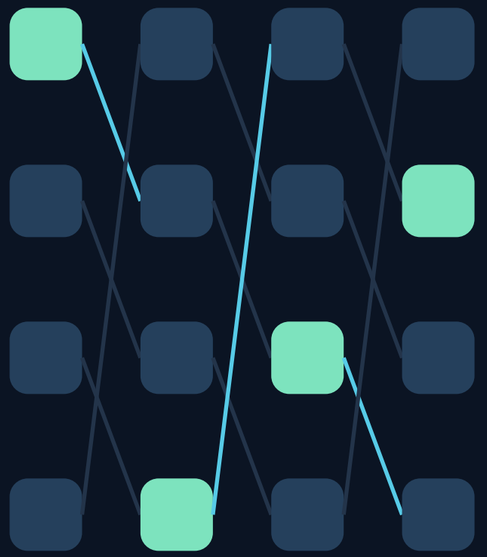
# Sleeping Machines

The next general-purpose substrate for machine intelligence.

One learner. Many forms of experience. Reusable intelligence, from representations to execution.

**Tero Keski-Valkama | Sole founder**

Raising €3M to scale research and engineering
Research-stage deep tech · 4 October 2026



**Capacity beyond activity**

# One substrate for learning, representing and executing intelligence.

Different forms of experience can teach reusable computations within the same learner.

**MANY FORMS OF EXPERIENCE**

## Observe + act

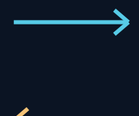
Tokens, sensors, typed tables and process traces. Synchronous samples and asynchronous events.



**ONE TRAINABLE SUBSTRATE**

## Learn + remember

Addressed messages, persistent memory and temporal races. End-to-end credit learns content, timing and unrealized alternatives.



**REUSABLE INTELLIGENCE**

## Compose + transfer

Transfer between perception, reasoning and action. Continual learning and self-design.

**The platform thesis: knowledge learned from one form of experience can become useful computation in another.**

Time and sleeps perform computation. Target: clockless learning and inference, capacity beyond activity, small data to large scale.

**First bridge: mixed-type tables through the temporal core. First transfer proof: joint learning, retained skills, matched work.**

# Nine public leaderboard wins, and language is next.

Official splits, sealed test sets scored once per run, the published state of the art as the bar. 40 of 41 independent runs ahead.

### EVENT PREDICTION

## 5 of 5

All five EasyTPP datasets won, and one configuration with no per-dataset tuning wins all five (24 of 25 seeds ahead).

### FAR SMALLER

## up to 15×

Smaller and cheaper than the state of the art (Retweet 1/15, Taxi 1/12, Amazon 0.29×), while more accurate.

### NEW DOMAINS

## 4 more

P19 sepsis (AUPRC 0.639 vs 0.583), PAM activity (0.978 vs 0.975), TGB trade (0.868 vs 0.863) and TGB wiki (0.835 vs 0.827).

### LANGUAGE

## lead grows

Tokenized text: ahead of a Kneser–Ney trigram (6.009 vs 6.537 nats/token), and further ahead at 4× the data.

**One configuration wins all five EasyTPP datasets; both temporal-graph benchmarks won at the first sealed attempt.**

# Public-leaderboard wins: timed event data (EasyTPP).

Public benchmark (ICLR 2024), official splits. Log-likelihood per event in nats: higher is better. Sealed test, 5 seeds.

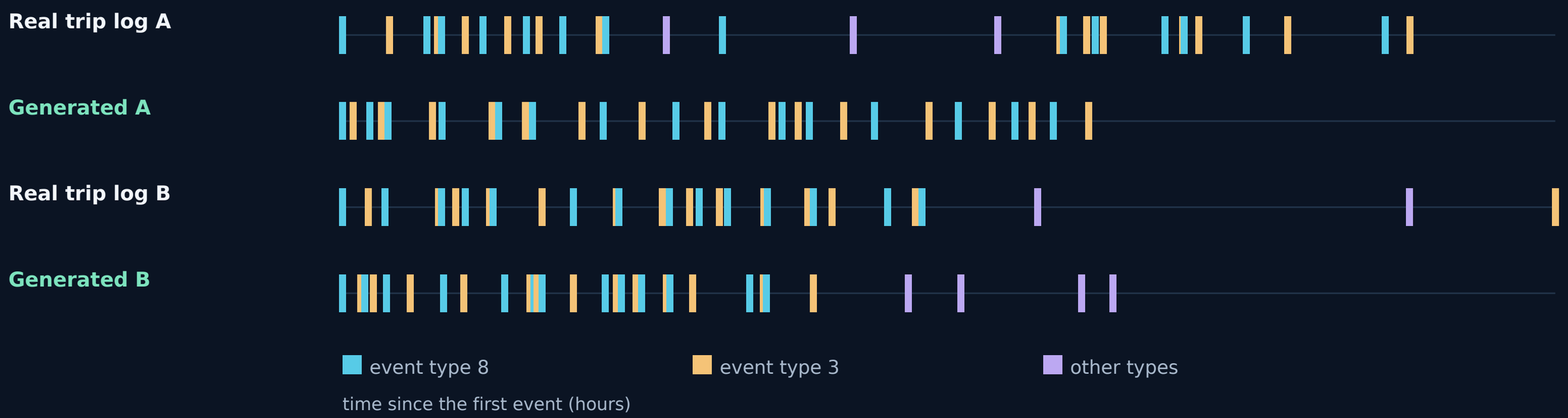
| Dataset | Best published | Ours | Result and compute we used |
|---|---|---|---|
| Taobao (shopping) | 1.318 | 1.399 ± 0.003 | Win · +0.081 nats/event at 0.92× the leader’s compute |
| Taxi (mobility), 5-seed mixture | 0.522 | 0.536 | Win · beats every published model at 0.41× its compute |
| Taxi, single model | 0.522 ± 0.004 | 0.525 ± 0.001 | Ahead on the mean at 1/12 of its parameters and compute |
| StackOverflow (Q&A activity) | −2.163 ± 0.009 | −2.144 ± 0.004 | Win · +0.019 nats/event at 1.26× the leader’s compute |
| StackOverflow, matched size | −2.163 ± 0.009 | −2.153 ± 0.005 | Win · all 5 seeds ahead at 1.0× the leader’s compute |
| Amazon (shopping reviews) | 0.781 ± 0.011 | 0.803 ± 0.001 | Win · all 5 seeds ahead at 0.29× the leader’s compute |
| Retweet (social) | −6.348 | −6.326 ± 0.001 | Win at 1/15 of S2P2's compute; all 5 seeds ahead |

**All five datasets won, each at about the state of the art’s compute or far less.**

Leaders: S2P2 on Taxi, StackOverflow, Amazon; IFTPP on Taobao; NHP on Retweet. Compute = per-event multiply-adds. Taxi reproduced on separate hardware.

# The same model also generates realistic event streams.

Taxi trips: real held-out sequences and sequences generated by racing the model's clocks from the same first event.




Over 200 held-out sequences: gap distribution KS distance 0.014 (naive reference 0.063); event-to-event transitions 0.024 total variation (naive 0.53).

**Exact sampling, no rejection: simulation, what-if analysis and synthetic data from the winning model.**

Naive reference: event types from overall frequencies, exponential gaps with the real mean. Single model, seed 0; examples drawn with a fixed seed.

# Public-benchmark win: ICU sepsis prediction (P19).

PhysioNet 2019, 38,803 ICU stays, 34 irregularly sampled channels, 4.2% positive. Five official splits, TEST once per split.

| Model | AUROC | AUPRC | Result |
|---|---|---|---|
| Best published (MTM, 2025) | 0.903 ± 0.020 | 0.583 ± 0.053 | State of the art on the official splits |
| Previous leaders (ViTST, GraFITi) | 0.892 / 0.891 | 0.531 / 0.566 | Published |
| Ours, 62,681 parameters | 0.916 ± 0.022 | 0.639 ± 0.039 | Win · AUPRC +0.056, beyond split spread |
| Ours without temporal memory | 0.900 ± 0.018 | 0.572 ± 0.037 | Below MTM on AUPRC |
| Trees on our statistics | 0.914 ± 0.020 | 0.618 ± 0.041 | Ours ahead on AUPRC, every split |

**The same temporal memory that won the event benchmarks wins clinical early warning.**

AUPRC is the clinically relevant metric at 4% prevalence. Temporal memory adds +0.072 AUPRC on every split. Reproduced on separate hardware.

# Beats the saved Transformer baselines at a fraction of the compute.

Wikipedia text (text8); same data and test set for all. Error: bits/char (bpc) or sMAPE %; lower is better.

| Comparison | Their error | Our error | Result and compute we used |
|---|---|---|---|
| vs Transformer, 4 layers, 4 passes (10M chars) | 1.908 bpc | 1.888 bpc | Better · 40% of its training, 18% of its inference compute |
| vs Transformer, 2 layers, 1 pass (10M chars) | 2.427 bpc | 1.955 bpc | Better · 96% of its training, 16% of its inference compute |
| vs LSTM-256 (10M chars) | 2.171 bpc | 1.955 bpc | Better at 88% of its inference compute (more training) |
| vs LSTM-512, 6 passes (10M chars) | 1.799 bpc | 1.888 bpc | Behind by 0.09; a wider model did not close it |
| vs best tuned LSTM (6 tuned runs) | 1.825 bpc | 1.888 bpc | Behind by 0.06; it used 92% of our compute |
| vs best tuned Transformer (4 tuned runs) | 1.996 bpc | 1.888 bpc | Better by 0.11 at ~equal compute (102%) |
| vs tuned Transformers (90M chars, ~2 PF) | 1.811 / 1.704 bpc | 1.783 bpc | Better than TF-256×4 at 1.06×; behind TF-192×4 |

**Wins vs both saved Transformers; a tuned small LSTM still leads. That gap is the next proof.**

Compute = estimated arithmetic operations. Baselines trained by us; 10 tuned runs at our two budgets complete, selected by validation. Scoreboard: evidence pack R1.

# Event-driven models aimed for parity. Ours beats small Transformers on compute.

Published event-driven and spiking language models, as reported by their authors, beside our completed result.

| Model | How it computes | Reported language result |
|---|---|---|
| EGRU on SpiNNaker2 (2023) | Clocked recurrent net; units fire events above a threshold | Matches LSTMs |
| SpikeGPT, 216M (2023) | Clocked binary spikes in a recurrent Transformer variant | “Competitive” with non-spiking models |
| Loihi 2 LLM, 370M (2025); sparse SSM (2026) | Conventional models made sparse/quantized; the chip supplies asynchrony | No accuracy loss / comparable to dense |
| Timing-coded LLM, 1.5B (2026) | Spike times carry values; closest to our races | “Clear gap” in language perplexity |
| Sleeping Machines (ours) | Learned arrival times choose which memories update; unchosen paths learn too | Beats two small Transformers at 40% of training compute; tuned LSTMs lead by 0.06 |

**To our knowledge, the first event-timed model to beat Transformer baselines on language with less compute, so far at small scale.**

Different datasets and scales; not a ranking. Ours: text8, 10M characters, small models, single runs, our baselines; LSTM-512 still ahead.

# Intelligence must compose across experience and execution.

A common substrate connects what a model learns with how and where it computes.

REPRESENTATION

## Meaningful inputs

Tokens, sensor events and typed comparisons enter through appropriate interfaces and act on shared persistent state.

LEARNING

## Reusable skills

End-to-end credit connects perception, memory, reasoning and action. Joint training tests transfer and retention.

EXECUTION

## Adaptive work

Learn timing and selective activity; recruit useful capacity for the task and the resources available.

**Small-data tables and large-data language are complementary tests of the same platform thesis.**

# One model family across today's architectural landscape.

Our approach combines strengths of today's main model types.

### Recurrent networks / Mamba

Exact member: learned delays over decaying memory (checked).

### Transformers

Exact member: races sample the softmax; delays give attention.

### Mixture of experts

Activate part of a large model.
We add private memory and timing.

## SLEEPING MACHINES / STATEFUL TEMPORAL PROGRAMS

One interface for content, timing, memory address and state, from single units to complete systems.
Target: computation through timing, sparse memory updates, small messages, and learning from unchosen paths.

### Statistics / retrieval

Counts and lookups of past data.
Combined with learned features.

### Adaptive memory

Models that adapt while in use.
With local, causal updates.

### Event-driven systems

Event-driven, neuromorphic systems.
We add trainable messages.

Mamba and attention containments are exact identities with numerical checks; other relationships are design relationships to be measured.

# Choose the right computation in each region.

One family can mix selective temporal programs with rich interaction and causal memory.

**01 / UNITS**

## Local programs

Each unit chooses what to remember, what message to send and when. Timing itself performs computation.

→

**02 / LAYERS**

## Adaptive work

A layer can activate one message or many: dense where needed, lean elsewhere.

→

**03 / COMPOSITIONS**

## Shared evidence

Learned features, counts and stored history can all inform the same prediction.

→

**04 / SYSTEMS**

## Same learner

The same learning rule works in batches or on a live stream, including learning after deployment.

**Counterfactual credit: learn from selected and unrealized routes**

**No global clock is required. Combined models, learning-after-deployment gains and chip energy are still milestones.**

# Messages race. Selected memories update. Alternatives learn.

A simple event loop connects representation, timing and sparse computation.

**01 / ENCODE**

## Signal + delay

A learned function produces a message and its arrival time from input and local memory.

→

**02 / SELECT**

## Temporal race

Arrival times select a receiver. Separate address keys and delivered values support sparse execution.

→

**03 / REMEMBER**

## Local state

Update the selected memory and propagate a new message into deeper computation.

→

**04 / LEARN**

## Alternatives

Use errors to improve message content and timing, including paths that were not selected.

**Counterfactual credit: learn from selected and unrealized routes**

**Implemented and trained today in ordinary software (CPU); no special hardware needed to run it.**

LEARNING ADVANCE

# Forward computation teaches more economical credit.

Real tokenized language: prospective credit accuracy and measured acquisition cost.

FRESH-DATA CREDIT

## 98.4% lower error

Eight fresh prefixes improve over static credit. A separate rank-eight block geometry test lowers error by 80.3% versus diagonal transport, with 27.1% more kernel work.

MATCHED ACQUISITION

## 2.09× faster

Versus exhaustive alternative learning. Context, route discovery, actual losses and factual correction included; 93.5% fewer supported operations.

CONNECTED CONTROLLER

## 77.8% less noise

Conditional gradient noise versus static credit in a connected controller with 37,377 parameters. Full factual correction preserves the exact mean gradient.

**Future-trained native optimizer improves quality in 9/9 seed/depth cases. Acquired-policy reuse tests whether this pays back in learning work.**

# If this scales, adaptation becomes a routine capability.

Conditional economic and industrial opportunity; the proof path is explicit.

ADAPT AS DATA CHANGES

## Continual learning

Useful models could adapt more often as processes, language and environments change. Prove quality and complete cost on an online workload.

LEARN LOCALLY

## Machines + devices

Smaller learning-state budgets could support industrial sensing, logistics and embodied systems. Measure memory, latency and energy on target hardware.

BUILD A BROADER PLATFORM

## Software + hardware

Reusable learned credit could serve many routes and modalities, with a sparse asynchronous execution path. Verify deeper-model scaling and cross-domain transfer.

**Investment case: make useful adaptation economical, then turn demonstrated savings into a deployable runtime.**

# One family wins across different kinds of data.

One construction: memories that evolve with elapsed time, sparse addressed writes, typed comparisons, races. The P19 classifier reuses the event model’s temporal memory layer, same code and size.

| Data type | Benchmark | Result |
|---|---|---|
| Generative event streams | EasyTPP (5 public datasets) | Win on all 5: Taxi, Taobao, StackOverflow, Retweet, Amazon |
| One shared core, four event domains | EasyTPP, joint training (development) | One memory matches or beats separate models on 3 of 4 |
| Irregular clinical records | P19 sepsis, official splits | Win: AUPRC 0.639 vs 0.583 |
| Wearable sensors (activity) | PAM, official splits | Win: accuracy 0.978 vs 0.975 (4 of 5 splits ahead) |
| Temporal interaction graphs | TGB trade and wiki leaderboards | Win on both: 0.868 vs 0.863 (NAVIS); 0.835 vs 0.827 (TPNet) |
| Anonymous interleaved process logs | FAS v1, 3 seeds | Win vs six generic controls (0.592 vs 0.559) |
| Tokenized language (GPT-2 BPE) | FineWeb vs Kneser–Ney trigram | Ahead: 6.009 vs 6.537 (3 seeds); lead grows at 4× data |
| Character language, 10M | text8 vs tuned baselines | Win vs Transformers at ≤ equal compute; LSTM leads |
| Temporal reasoning, few examples | Structured event tasks | 99.7% vs Transformers 33–41% |
| Mixed-type tables | Synthetic; banknote | Learns the interaction; trees lead on real data |

**S2P2, the event-benchmark leader, is shown on one task type. We win on several.**

Large-scale language (90M characters) and real tables remain behind; the tokenized-language reference is a count model, a published small Transformer is next.

# Respect types. Race decisions. Compose learned state.

Meaningful comparisons become events before evidence enters a common neural representation.

RESPECT THE FIELD

## Typed comparisons

Compare a quantity with a threshold in its units. Test category membership, Boolean conditions and missingness explicitly. Category codes need no invented distance or order.

LEARN THE DECISION

## Temporal races

Comparison outcomes feed computational races and selected memory writes. Counterfactual consequences teach routes. Learned thresholds require branch, timing and factual-path credit; that extension is the next test.

COMPOSE THE EVIDENCE

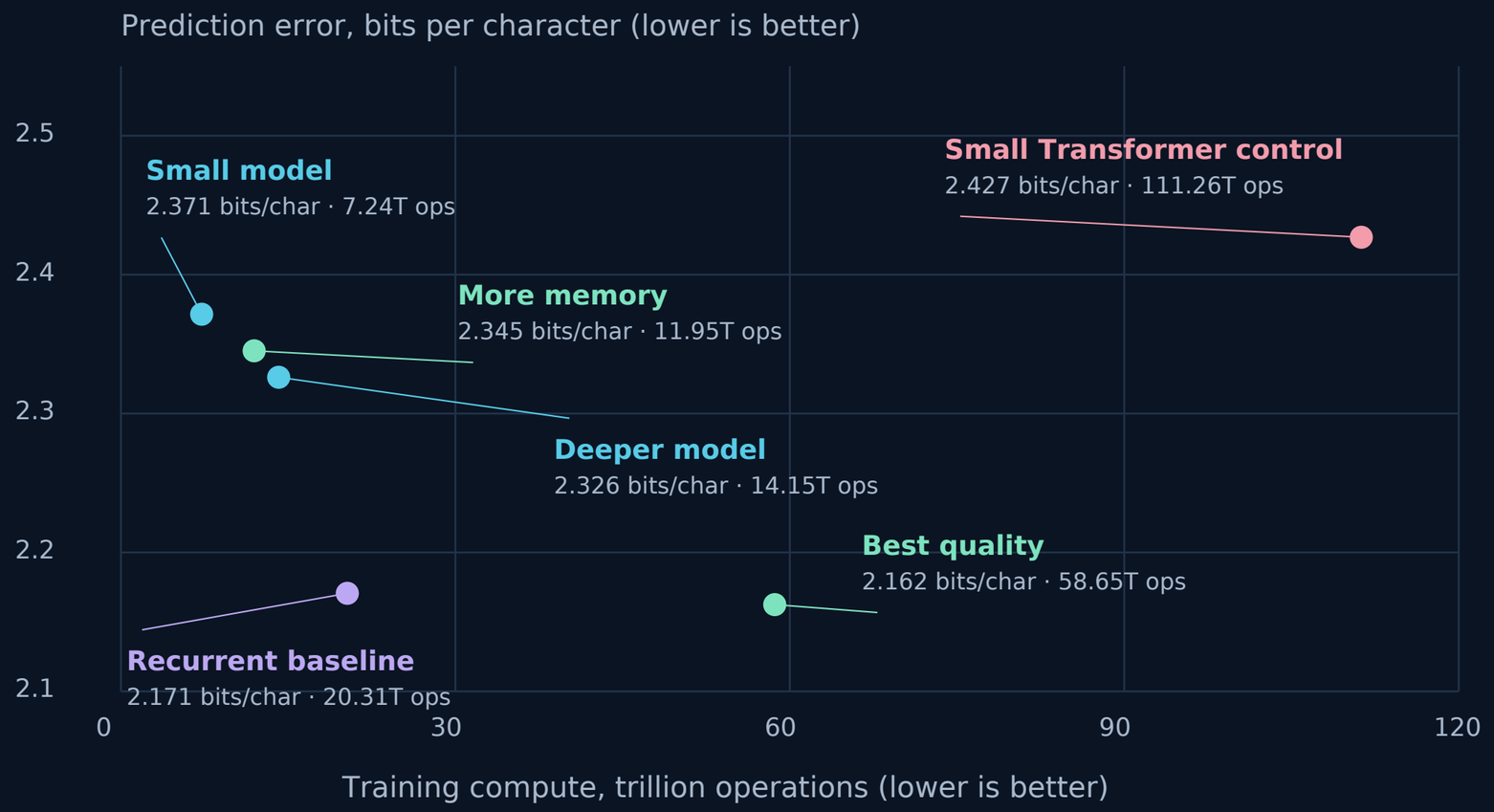
## Deep shared state

After a typed interface, learned vectors can mix meaningfully in persistent representations. The same event construction can connect tables, tokens, process traces and actions.

**Typed semantics + conditional selection + deep temporal learning. First mixed-type fit:100% synthetic DEV accuracy; fixed predicates.**

# Single pass: error versus training compute.

Each model sees the 10M training characters once. Lower and further left is better. Slide 2 adds multi-pass runs.




**BEST SINGLE PASS**

## 2.162 bpc

Edges LSTM-256 (2.171), with 2.9× its training compute.

**TRAINING COMPUTE**

## 15.4× less

Small model vs 2-layer Transformer: less compute and lower error.

Baselines trained by us on the same data and test set. Compute is estimated arithmetic (trillion operations), not measured energy or cost. Single runs.

# Start with low-batch language inference on existing hardware.

Product hypothesis for local and on-premise text applications; no customer interest reported yet.

WHO WOULD EVALUATE IT

## Platform teams

Teams deploying small language models under latency, memory or cost constraints. Demand validation is a next step.

WHAT WE WOULD SUPPLY

## Model + runtime

A trained model, sparse inference runtime and a reproducible quality-and-cost evaluation on the intended workload.

WHY THEY WOULD ADOPT

## Verified savings

Adoption depends on equivalent useful quality, service reliability and a measured reduction in complete deployment cost.

**Commercial gate: target ≥20% full-system cost reduction at matched quality on one selected workload.**

# Software proves the advantage. Hardware multiplies it.

One research program, with staged routes to deployment.

01 / MODELS + RUNTIME

## First product

Validate the language workload on existing hardware. License a model/runtime with auditable deployment economics.

02 / MEMORY-LOCAL ACCELERATORS

## 1.7–5.8× less data

Modeled per character: 1.7× fewer memory bytes than the best tuned Transformer and 5.8× fewer than the saved one, both beaten. 10× more memory adds 5% traffic. Tuned LSTMs: parity.

03 / CLOCKLESS SILICON

## Work only on events

No global clock: idle parts draw almost no power, answers arrive as soon as they are ready, and cool chips allow dense 3D stacking. Also the missing model for neuromorphic chips.

**The €3M round funds scale and a hardware cost model of the trained network; silicon follows measured proof.**

# A useful alternative could strengthen a hardware ecosystem.

Potential evaluation and licensing routes; no partner interest or relationship is claimed.

AMD

## Model + platform

Its ~$665M Silo AI acquisition and Liquid AI collaboration show the value of model/software capability on AMD hardware.

INTEL

## Event execution

Hala Point and Loihi 2 provide a relevant research environment for expressive event models and continuous learning.

GOOGLE

## Full-stack design

TPU model–hardware co-design creates a demanding evaluation venue for quality, memory and execution improvements.

**Partner proposition: demonstrate a differentiated workload advantage that improves platform adoption.**

# Price the runtime against verified customer value.

Illustrative annual license economics; assumptions to test with an evaluation partner.

| Per-customer assumption | Calculation | Annual amount |
|---|---|---|
| Eligible workload cost | Existing deployment cost we can actually influence | €1.00M |
| Gross savings | 20% complete-cost reduction at matched quality | €200k |
| Runtime license | 20% of verified gross savings | €40k |

Customer retains €160k/year before migration and other costs. At the same assumed price, 25 deployments would generate €1M annual license revenue.

**Validate pricing through matched quality, reliable operation and measured customer benefit.**

# AI research and software architecture in one founder.

Tero Keski-Valkama · Sole founder · Public career and invention record.

## Tero Keski-Valkama

### Sole founder

Experience spans applied ML, multimodal foundation models and software architecture. The round adds specialist capacity around the founder.

| | |
|---|---|
| **2012–2018** | **Cybercom · Software Architect**<br>Historical role in public professional profile. |
| **2018–2022** | **HERE · Lead ML & AI Engineer**<br>Self-reported research and applied ML leadership. |
| **2022–2025** | **Alloy.ai → Kaiko.ai**<br>Staff engineering; multimodal foundation-model work. |
| **2019–2025** | **35 patent publications · Inventor**<br>US and EP patents; assignee HERE Global. |

Historical roles are self-reported; inventorship is documented. Planned hiring: learning research, runtime engineering, evaluation and measurement.

# Build defensibility across the model and its execution.

The strongest potential asset is an integrated stack that is difficult to reproduce and deploy.

MODEL + LEARNING

## Research know-how

Timing, state and alternative-credit methods, supported by retained ablations and numerical contracts.

RUNTIME + HARDWARE

## Execution know-how

Sparse kernels, memory management, measurement and hardware mapping can reinforce the model advantage.

RIGHTS + PATENTS

## Company assets

Secure company ownership. Assess technical inventions for selective patent filings; protect confidential implementation know-how.

**Funding builds company-owned IP: ownership review, selective patent filings and confidential know-how.**

# Three gates determine whether this becomes a business.

Each gate has a falsifiable result and a consequence for further spending.

01 / QUALITY AT SCALE

## Replicate

At least three runs per result, fixed held-out tests and tuned baselines. Compare quality and total compute; publish failures too.

02 / COMPLETE ECONOMICS

## Measure

Show the fast sparse runtime gives the trained model’s answers. Measure serving cost, latency, memory and energy on one workload.

03 / ADOPTION

## Evaluate

Identify a use case and an evaluator willing to test it. Confirm useful quality, integration effort and willingness to pay.

**If a gate fails, diagnose or narrow the workload before committing more compute or new silicon.**

# The next 18 months produce evidence investors can underwrite.

Proposed milestones measured from funding; spending advances with completed gates.

**MONTHS 0–3**

## Team + proofs

Hire the initial team. Resolve IP ownership and review priority filings. Verify the trained sparse runtime; repeat larger-data results.

**MONTHS 3–6**

## One workload

Run tuned controls and measure total cost, latency, memory and energy. Select a workload only if quality and economics hold.

**MONTHS 6–12**

## External trial

Seek independent reproduction and prospective customer evaluations. Test pricing, integration effort and willingness to pay.

**MONTHS 12–18**

## Product + scale

Turn a validated workload into a deployable runtime. Expand models and assess hardware mapping for the next financing stage.

**Next-round objective: replicated quality, a measured cost advantage and a credible evaluation pathway.**

USE OF FUNDS

# €3M funds an 18-month research and engineering program.

Team, staged compute, hardware feasibility and IP protection, including selective patent filings.



| | |
|---|---|
| Team: research + systems | **€1.35M \| 45.0%** |
| Compute + independent replication | **€0.90M \| 30.0%** |
| Hardware feasibility + measurement | **€0.25M \| 8.3%** |
| IP, legal, operations | **€0.20M \| 6.7%** |
| Contingency reserve | **€0.30M \| 10.0%** |

**€2.70M operating plan + €0.30M reserve · Planned operating burn averages €150k/month.**

Budget assumptions: six average FTE including founder at €150k/year fully loaded; compute and hiring quotes remain to be obtained.

# Raise €3M to establish a commercially valuable platform.

Proposed priced round; terms remain subject to company formation and investor diligence.

**CAPITAL TO RAISE**

## €3M

18 months of research, systems engineering and staged validation.

**PROPOSED PRE-MONEY**

## €50M

A premium research-stage proposal for the model/runtime/substrate opportunity.

**INITIAL OWNERSHIP**

## 5.66%

€53M post-money, before option-pool changes, preferences and later dilution.

The proposal prices a model/learning family, sparse runtime and future chip architecture that already beats Transformer baselines at a fraction of the compute. A confirmed 90M-character win and a NeuroBench first place would justify a higher next price (the €100M scenario in the appendix).

**Valuation sensitivities and conditional exit scenarios are in the appendix.**

# Build the model. Prove the economics. Open the hardware path.

€3M proposed raise · €50M proposed pre-money · Private investor discussion.

# €3 million

Already beating Transformer baselines on compute.
This round tests whether the advantage scales.

**01** Replicated language quality and learning mechanisms.

**02** A deployable runtime with measured system economics.

**03** External evaluation and a credible path to hardware.

Tero Keski-Valkama · Private investor diligence available by arrangement